

GERT8000 REVISION NOTES

Semaphore Signals & Signalling Systems

Based on classroom session notes | Cross-referenced to GERT8000 Rule Book

1. STOP SIGNALS

Appearance

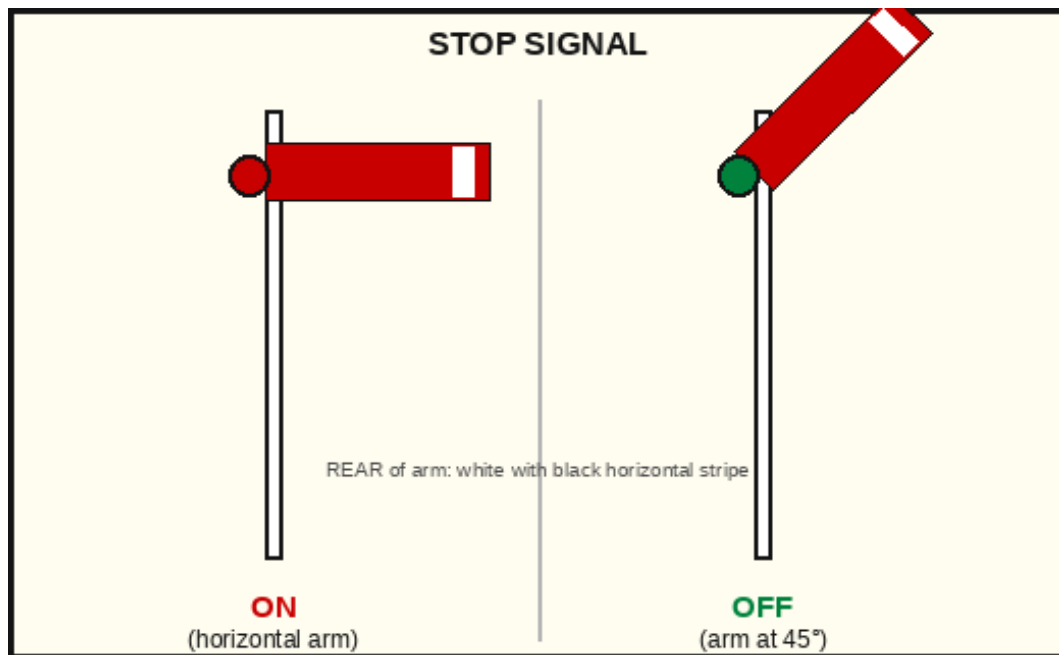


Fig. Stop signal — ON (danger) and OFF (clear)

Signal	Position / Aspect	Meaning	Light
Stop Signal	ON (horizontal arm)	DANGER – Do not pass	
Stop Signal	OFF (arm at 45°)	PROCEED – Line is clear	
Stop Signal (rear)	—	White with black horizontal stripe	

Upper & Lower Quadrant

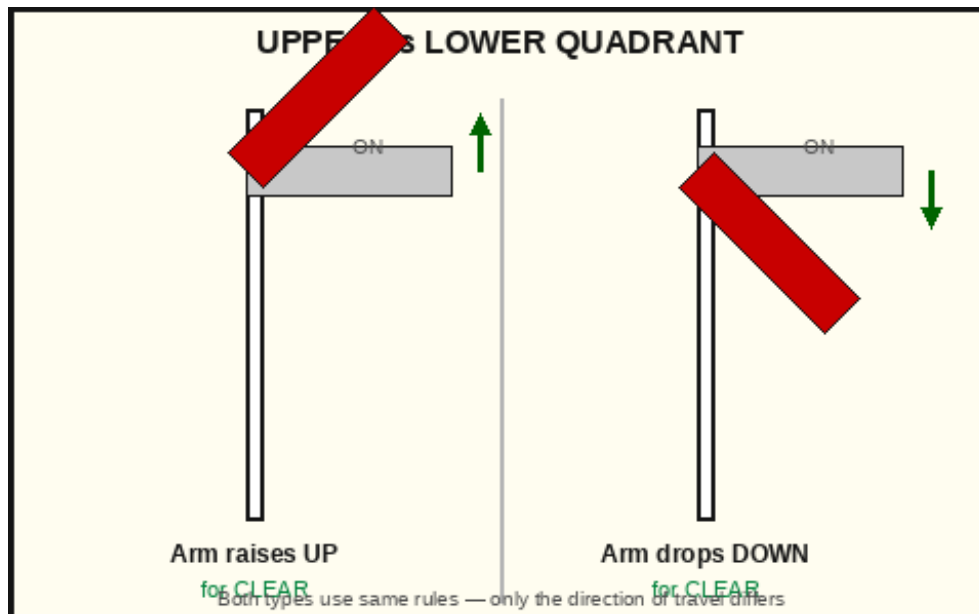


Fig. Upper quadrant (arm raises) vs Lower quadrant (arm drops)

- Upper Quadrant: arm raises to 45° for clear (most modern semaphore areas).
- Lower Quadrant: arm drops to 45° for clear.

Both types are in use. The Rule Book applies equally to both.

Multiple Stop Signals (Home Signals)

- A block section may have up to 4 home signals.
- If 2 home signals: the furthest from the signal box = Outer Home; the nearest = Inner Home.
- If 2 signals on the same post, you must obey both.

GERT8000-S7 §1.1b: You must obey each signal which applies to the movement of your train.

Stopped/Nearly Stopped at a Semaphore

⚠ If you have stopped or nearly stopped at a semaphore signal at danger and it then clears, you must be prepared to stop at the NEXT stop signal worked by the same signal box.

GERT8000-S7 §1.6 | Exception: does not apply to signal controlling entrance to an IBS section.

Signal Not Showing Correctly

- Treat a stop signal as AT DANGER.
- Treat a distant signal as AT CAUTION.

GERT8000-S7 §1.5: Applies if arm not showing, wrong aspect, or semaphore not showing correctly.

2. DISTANT SIGNALS

Appearance

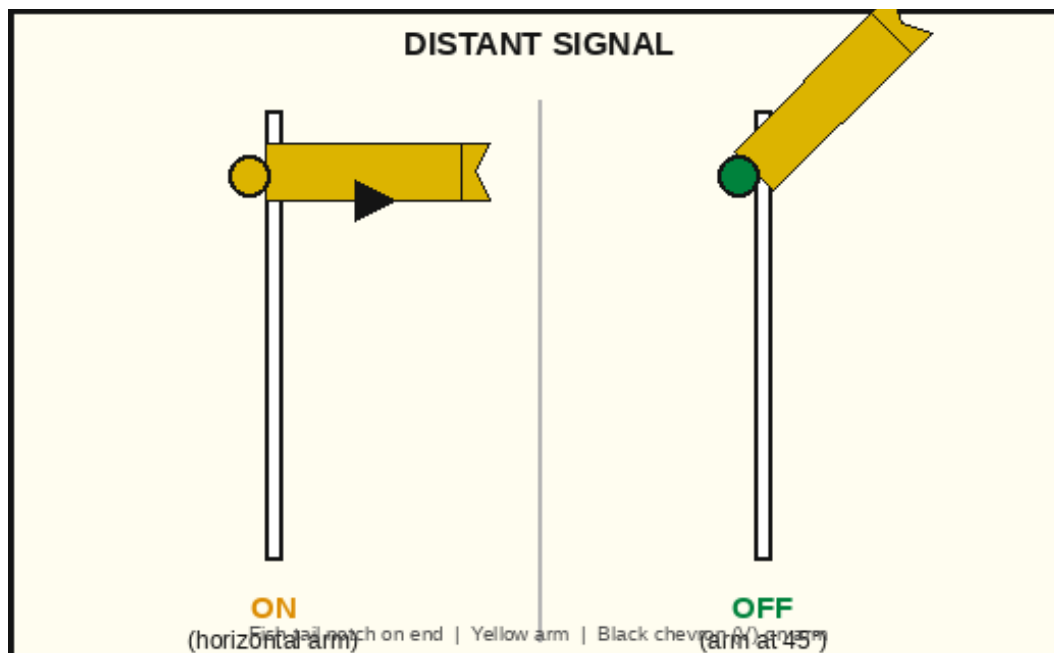


Fig. Distant signal — ON (caution) and OFF (clear). Note fish-tail notch and black chevron.

Signal	Position / Aspect	Meaning	Light
Distant Signal	ON (horizontal arm)	CAUTION — Be prepared to stop at a stop signal ahead	
Distant Signal	OFF (arm at 45°)	CLEAR — All associated stop signals are off	
Distant Signal (front)	—	Yellow arm with fish-tail notch; black chevron (V) on arm	
Distant Signal (rear)	—	Fish-tail shape, white with black chevron	

Key Rules

- If DISTANT is ON: be prepared to stop at any of its associated stop signals.
- If DISTANT is OFF: all associated stop signals MUST also be off.

⚠ The distant signal gives advance warning only — it is NOT a stop signal. Never pass a stop signal at danger on the authority of the distant.

GERT8000-S7 §1.5: Treat a distant signal showing incorrectly as AT CAUTION.

Fixed Distant Signal

- Painted on a board (not a moving arm) — permanently indicates caution.
- Can be placed before a buffer stop or on approach to a terminal area.
- Always treat as AT CAUTION.

Outer Distant Signal

- Fitted ahead of the normal distant where block sections are unusually long.
- Gives an earlier warning before the standard distant signal.

3. ABSOLUTE BLOCK WORKING & SIGNAL LAYOUT

Signal Order (approaching a station/box)

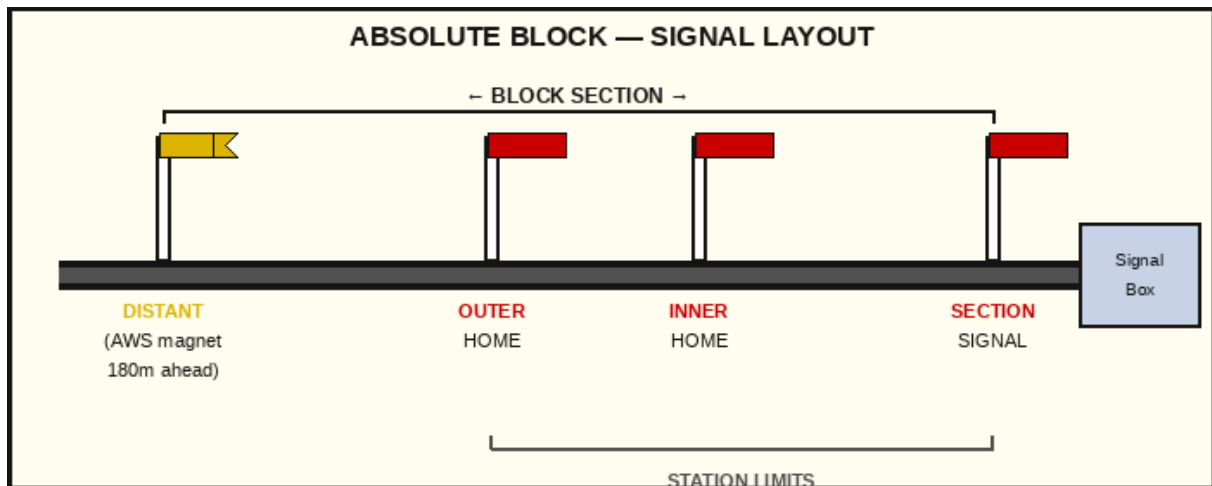


Fig. Absolute block signal layout — DISTANT → HOME → SECTION, with station limits

DISTANT → HOME (Outer/Inner) → SECTION

The BLOCK SECTION runs from the previous signal box's section signal up to the next box's home signal.

Station Limits

- Defined as: from the FIRST HOME SIGNAL up to (and including) the SECTION SIGNAL.
- All movements within station limits are the responsibility of the signaller for that box.

Service Braking Distance

- The distance from the distant signal to the home/section signal — must be sufficient for a train to brake to a stop.

Clearance Point / Overlap

- The distance of track beyond a home signal that must be clear before the previous signal is returned to danger.
- Protects against a train over-running slightly after the signal clears.

Block Instrument

- Used by signallers to communicate train movements between adjacent signal boxes.
- Three standard indications: LINE BLOCKED, LINE CLEAR, TRAIN ON LINE.

► **NOT VERIFIED:** this Block Instrument entry, including the names of the three indications, could not be matched against any of the uploaded rule book material (GERT8000-S7, SS2, S4, S5, RS521). Please double-check this against your classroom notes or confirm before relying on it.

Fail-Safe Design

⚠ **Semaphore signal cables FAIL SAFE to ON (danger). A broken cable puts the signal to danger automatically.**

Distant on Same Post as Section Signal

- The NEXT signal box controls the distant arm.
- The PREVIOUS signal box controls the section signal arm.

Fringe Signal Boxes

- Located at the boundary between semaphore (absolute block) territory and colour-light territory.
- Special working arrangements apply at the fringe; drivers should be aware of the transition.

4. AUTOMATIC WARNING SYSTEM (AWS)

Location

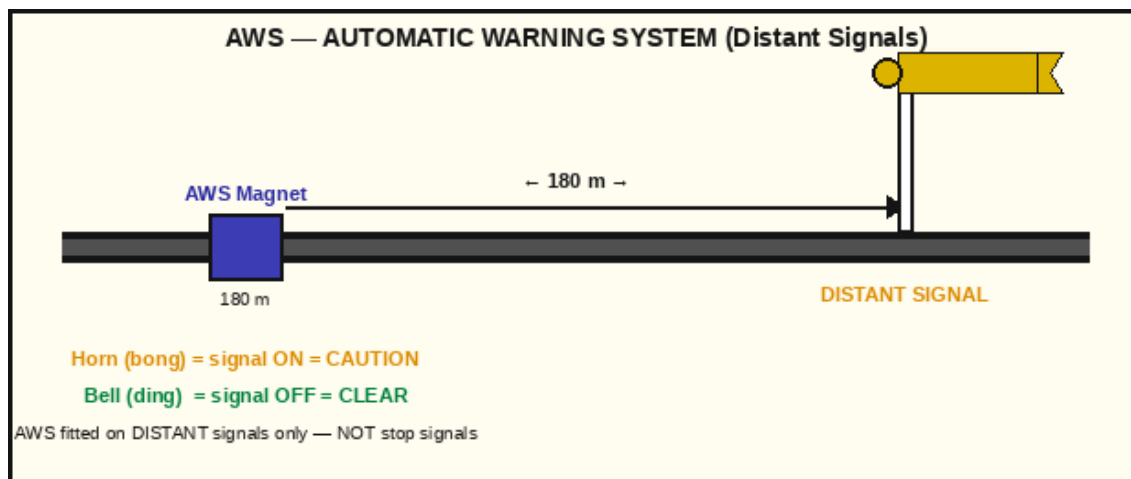


Fig. AWS magnet position — 180 m in advance of distant signal

- AWS track magnet fitted to DISTANT signals only (not stop signals).
- Magnet is positioned 180 m in advance of the distant signal.

Indications

Signal	Position / Aspect	Meaning	Light
AWS	Horn (bong)	Warning — distant signal is ON (caution). Must cancel immediately.	
AWS	Bell (ding)	Clear — distant signal is OFF. Acknowledge and proceed.	

⚠ **You must IMMEDIATELY cancel each AWS warning. Failure to cancel within the allowed time causes an automatic brake application. If brakes apply: stop the train, tell the signaller.**

GERT8000-S7 §5.1: Cancel AWS warning indication immediately.

AWS Warning with Distant Showing CLEAR

⚠ **If the distant signal is SHOWING CLEAR but you receive an AWS WARNING (horn), you must treat the distant signal as being AT CAUTION.**

GERT8000-S7 §5.2 | Exception: if signal changed to clear after you passed the magnet, or a warning board is positioned at the signal.

Unexpected AWS Warning

- If you receive a warning with no AWS track equipment present: proceed normally, report to signaller at earliest opportunity.

GERT8000-S7 §5.3

5. INTERMEDIATE BLOCK SIGNALS (IBS)

Purpose

- Divides a long block section into two shorter sections, increasing line capacity.
- Located between two signal boxes (not controlled by either box directly).

Identification

- White square sign with a vertical black line on the signal post.

Outer Distant for IBS

- Where the block section before an IBS is long, an outer distant signal is provided.
- This outer distant is before (in advance of) the IBS.

Level Crossings within the Block Section

- A gatekeeper-controlled level crossing may have its own distant and stop signal.
- These crossing signals are NOT part of the normal block signalling system.
- They control the crossing independently of the block section.

6. SUBSIDIARY SIGNALS & PERMISSIVE WORKING

Permissive Working

- Allows two trains to occupy ONE block section simultaneously.
- Authorised by a SUBSIDIARY SIGNAL fitted on the same post as the stop signal.
- Only proceed on the subsidiary if the main stop signal aspect is not showing proceed.
- When proceeding on a subsidiary: PROCEED AT CAUTION.

Subsidiary Signal Types

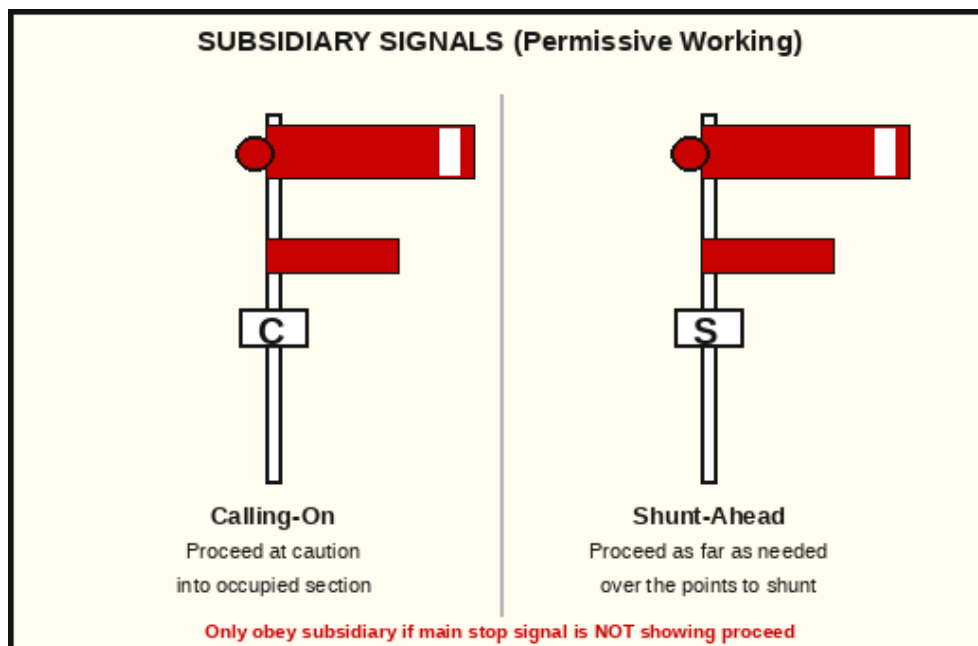


Fig. Calling-On (C) and Shunt-Ahead (S) subsidiary signals below main stop arm

Signal	Position / Aspect	Meaning	Light
Subsidiary Signal	Letter C below arm	CALLING-ON — same meaning as normal subsidiary; proceed at caution into occupied section	
Subsidiary Signal	Letter S below arm	SHUNT-AHEAD — train may proceed as far as needed over the points to complete the shunt	

⚠ Passenger trains must NOT proceed on the authority of a semaphore SHUNTING SIGNAL. They may proceed on a subsidiary signal only when entering a permissive platform line.

GERT8000-S7 §3.1

Route Indication with Subsidiary

- If a subsidiary signal clears but the normal route indication is NOT shown:
- → Proceed at caution, be prepared to stop before any obstruction.

GERT8000-S7 §3.2

7. SEMAPHORE SHUNTING SIGNALS

Types of Shunting Signal

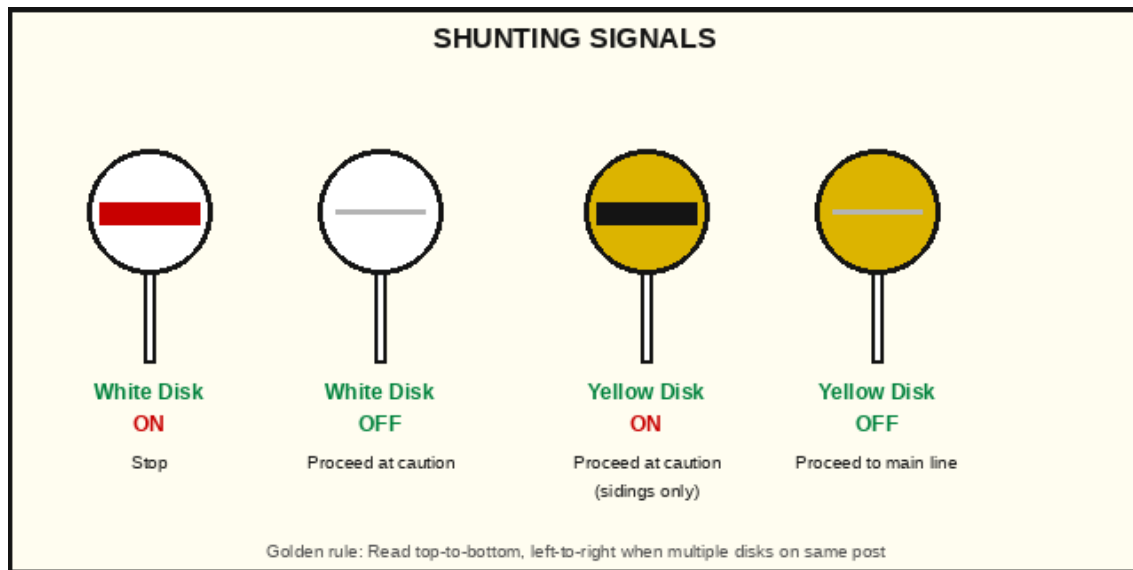


Fig. Shunting signal types — white disk ON/OFF and yellow disk ON/OFF

Signal	Position / Aspect	Meaning	Light
Shunting Disk	White disc with red horizontal bar — ON	Stop — do not pass	
Shunting Disk	White disc with red bar — OFF (cleared)	Proceed at caution over the points	
Miniature Semaphore	Small stop arm with vertical white stripe — ON	Stop	
Miniature Semaphore	OFF (arm lowered)	Proceed at caution	
Yellow Shunting Disk	ON (showing)	Proceed at caution within the sidings	
Yellow Shunting Disk	OFF (cleared by signaller)	Proceed from sidings onto main line	

Golden Rule — Reading Shunting Signals

⚠ **Read shunting signals TOP to BOTTOM, LEFT to RIGHT. Where 3 disks are displayed: top disk = left line, middle disk = middle line, bottom disk = right line.**

After Shunting — Returning to Journey

- After a shunting movement on the authority of a shunting/shunt-ahead signal:
- 1. The movement must return to the APPROACH SIDE of the signal.
- 2. The signal must display the appropriate PROCEED aspect/indication before continuing the journey.

GERT8000-S7 §3.3

8. LIMIT OF SHUNT

What it is

- A defined point beyond which a shunting movement must NOT pass without permission.
- Three types of sign are used in the Rule Book to indicate a limit of shunt.

► **NOT VERIFIED:** the claim that exactly “three types of sign” indicate a limit of shunt could not be matched against the uploaded rule book material. The underlying rule below it (must not pass without the signaller’s permission) IS confirmed at GERT8000-SS2 §5.5. Please double-check the “three types” detail against your classroom notes or confirm before relying on it.

- Passing beyond the limit of shunt without permission = a SPAD (Signal Passed at Danger).

⚠ **You MUST NOT allow any part of the movement to pass a limit of shunt signal or indicator unless the SIGNALLER has given permission.**

GERT8000-SS2 §5.5

Wrong-Direction Movement Beyond Home Signal

- A wrong-direction shunting movement must NOT go beyond a home signal without the signaller's permission.
- If there is a falling gradient towards the next signal box, the movement must not be made unless:
 - a) the automatic brake is working throughout the train, OR
 - b) the locomotive is at the end nearer to the next signal box.

GERT8000-SS2 §5.6

9. ROUTE INDICATORS (DIVERGING ROUTES)

Signal	Position / Aspect	Meaning	Light
Stepping Route Indicator	Tallest position	Main line (or principal route)	
Stepping Route Indicator	Lower positions	Diverging routes in sequence — lower = more divergent	
Stacking Route Indicator	Position varies	Each indicator lit = specific diverging route	
Route Indicator (theatre)	Letter/number displayed	Specific route shown alphanumerically	

Where two signals are on the same post, obey both signals — apply the most restrictive aspect.

10. CO-ACTING SIGNALS

► **NOT VERIFIED:** this entire Co-acting Signals section has no citation, and no matching text was found anywhere in the uploaded rule book material (GERT8000-S7, SS2, S4, S5, RS521) or the Shunting PPTX. Please double-check this against your classroom notes or confirm before relying on it.

- Two semaphore arms on the SAME post, both showing the same aspect simultaneously.
- Fitted to improve sighting — usually where one arm would not be visible from the full approach distance.
- Both arms move together (they co-act). Treat as a single signal.

11. WHITE DIAMOND INDICATORS

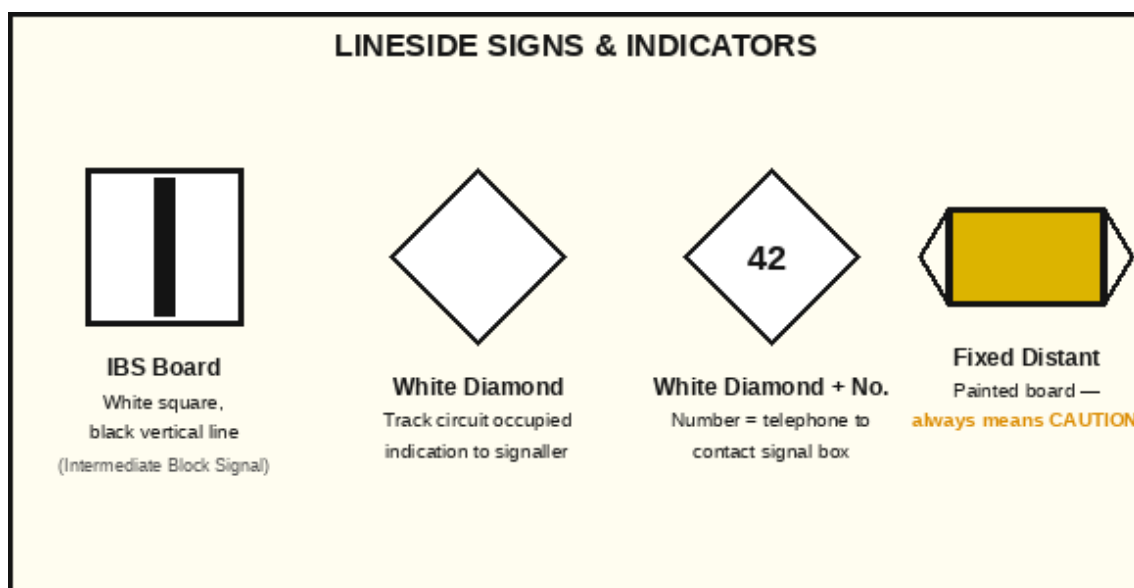


Fig. IBS board, white diamond, white diamond with telephone number, fixed distant board

Signal	Position / Aspect	Meaning	Light
White Diamond	Plain	Confirms the presence of a train or shunting movement is indicated to the signaller by track circuit.	
White Diamond + number	Number displayed	A telephone is not provided at the sign itself, but the presence of the train or shunting movement is indicated to the signaller. If GSM-R is not available, the signaller may be contacted using the telephone number shown on the plate.	

RS521 §11.3: White diamond sign (plain, or with a telephone number) — a telephone is not provided, but the presence of the train or shunting movement is indicated to the signaller. Where a number is shown and GSM-R is unavailable, that number may be used to contact the signaller. (GERT8000-S4 §2.2 covers a separate procedure — using another telephone if you must leave the cab during an emergency or when the line is blocked — and is not the source for the sign's meaning.)

Note: On lines with no block working (non-monitored lines), no white diamond will be present.

12. REPORTING SIGNALLING IRREGULARITIES

⚠ You must tell the signaller IMMEDIATELY (stopping the train specially if necessary) if you become aware of any signalling failure or irregularity — including a semaphore signal not showing correctly.

GERT8000-S7 §7.1

What to report immediately

- Semaphore signal not showing correctly (arm stuck, wrong position, etc.).
- White light showing instead of red, yellow or green.
- No signal shown when there should be one.
- Irregular aspect sequence.
- Any failure or irregularity in the working of AWS.

Signals that are not in use

- A semaphore signal no longer in use will have a WHITE CROSS placed over it.
- A shunting disk no longer in use will be covered over.

Do not act on a signal marked out of use.

Reporting — Form RT3185

- When a signal/AWS failure is reported, both driver and signaller must complete Form RT3185.

GERT8000-S7 §7.8a

- Exception: you do not need to complete RT3185 if the fault is clearly a right-side failure (or you can explain why it is one and are fully aware of the circumstances). You must still tell Operations Control and make a suitable entry in the Train Register. If the signaller can identify or explain the fault as a right-side failure, you do not need to complete RT3185 immediately — complete it at the first convenient opportunity instead.

GERT8000-S7 §7.8c

13. QUICK REFERENCE — KEY RULES AT A GLANCE

Situation	Rule / Action	Reference
Stop signal ON (horizontal)	STOP — do not pass	S7 §1.1b
Stop signal OFF (45°)	PROCEED — line is clear	S7 §1.1b
Distant signal ON	Caution — be prepared to stop at stop signals ahead	S7 §1.1b
Distant signal OFF	Clear — all associated stops are also off	S7 §1.1b

Signal not showing correctly	Stop signal → treat as DANGER; Distant → treat as CAUTION	S7 §1.5
Stopped at semaphore; signal then clears	Proceed but be prepared to stop at NEXT stop signal (same box)	S7 §1.6
AWS horn received	Cancel IMMEDIATELY; obey caution; prepare to stop at home signal	S7 §5.1
AWS horn; distant shows clear	Treat distant as AT CAUTION	S7 §5.2
Subsidiary signal clears; no route shown	Proceed at caution; prepared to stop at any obstruction	S7 §3.2
Passenger train + shunting signal	Must NOT proceed on shunting signal authority	S7 §3.1
After shunting on shunt signal authority	Return to approach side of signal before continuing journey	S7 §3.3
Limit of shunt indicator	Must NOT pass without signaller's permission (= SPAD if exceeded)	SS2 §5.5
Wrong-direction move beyond home signal	Must have signaller's permission first	SS2 §5.6
Semaphore not showing correctly	Report to signaller IMMEDIATELY (stop train if necessary)	S7 §7.1
Broken semaphore cable	Signal FAILS SAFE to ON (danger)	General

These revision notes are a study aid. Always refer to the current issued GERT8000 Rule Book modules for authoritative rules.